Synthetic Cold Splits Are Not Enough: Distribution-Shift Benchmarking for Drug-Target Interaction Prediction

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Abstract

Drug-target interaction (DTI) papers often use synthetic cold-start splits as proxies for generalization, but it remains unclear whether conclusions drawn from these splits transfer to real distribution shifts. We present a benchmark-first case study showing that, within an expanded five-model panel, rankings are highly sensitive to how the shift is defined. The panel contains a fully specified fixed-feature `base` encoder, a learner-control `RandomForest` baseline on the same Morgan+k-mer inputs, two recent pooled-LM baselines (`DTI-LM` and `HyperPCM`), and a `GraphDTA-style` graph-family probe. On a 3-seed synthetic reference panel built from `BindingDB_Kd`, the winner depends strongly on the split: `RF` is strongest on `unseen_drug` (0.7339 ± 0.0051 AUPRC) and `blind_start` (0.4512 ± 0.0373), while `base` remains strongest on `unseen_target` (0.5224 ± 0.0602). On the promoted temporal/provenance benchmarks, the ranking changes again. Over 3 seeds, `RF` becomes strongest on both `BindingDB_patent / patent_temporal` (0.8234 ± 0.0012) and `BindingDB_nonpatent_Kd / nonpatent_temporal` (0.7201 ± 0.0012), with paired bootstrap intervals remaining strictly positive against both `base` and `DTI-LM`. In contrast, the `GraphDTA-style` probe is competitive but does not dominate any promoted axis. These additions strengthen the benchmark claim rather than weakening it: ranking instability is not confined to pooled-LM comparisons, and even a classical learner on fixed handcrafted inputs can become the strongest system under both synthetic and temporal evaluation. A pooled year-band and overlap-bucket diagnosis further shows that temporal/provenance OOD is structured rather than monolithic. We package the split manifests, benchmark resources, and standalone evaluator for reuse. Within the studied model panel and benchmarks, these results show that synthetic cold splits and real OOD shifts should not be treated as interchangeable evaluation settings for DTI.

1 Introduction

Predicting drug-target interactions for unseen molecules or proteins is a central goal in computational drug discovery. Because truly prospective evaluation is difficult and expensive, many recent studies report held-out drug, target, or blind splits on datasets derived from BindingDB or DAVIS as proxies for inductive generalization [1, 2, 3, 4, 5, 6, 7, 8]. These protocols are useful, but they also encourage a strong implicit assumption: if a method performs well on one cold-start split, then it is broadly robust to the kinds of shift encountered in deployment.

That assumption is weaker than it appears. Different split definitions remove different forms of overlap, and they may favor different model families. A pooled-LM baseline that looks strong under one synthetic exclusion rule may no longer be the most reliable option once the benchmark shifts in time, provenance, or assay composition. If these ranking changes occur even within a compact studied panel, benchmark design is not a technical footnote. It becomes part of the scientific claim.

This paper studies that question directly. Rather than proposing another universal architecture, we ask whether the ranking of an expanded DTI model panel is preserved across synthetic cold-start regimes and real distribution shifts. The goal is to test ranking stability under benchmark shift in a controlled case

study, not to declare a universal field-wide leaderboard. Our panel begins with three representative systems: a fixed-feature MLP reference model (`base`) and two recent external literature baselines, `DTI-LM` and `HyperPCM`. We then add two deliberately different controls: a `RandomForest` classifier trained on the same Morgan+k-mer representation as `base`, and a `GraphDTA-style` graph-family baseline. Even on the synthetic reference panel, rankings are already heterogeneous: `RF` is strongest on synthetic `unseen_drug` and `blind_start`, while `base` remains strongest on synthetic `unseen_target`.

We then contrast this synthetic picture with real temporal/provenance benchmarks built from BindingDB patent and non-patent time-based splits. The result is another ranking change. The synthetic panel does not point to a single dominant system, but the promoted temporal benchmarks strongly favor `RF`. Paired bootstrap confidence intervals support these temporal comparisons: `RF - base` and `RF - DTI-LM` remain positive on both `patent_temporal` and `nonpatent_temporal`. An earlier-cutoff variant of the patent benchmark (`patent_temporal_v2017`) had already preserved the same temporal ordering within the smaller panel, so the expanded results strengthen rather than overturn the original benchmark diagnosis.

To move beyond leaderboard-style reporting, we also diagnose the promoted temporal benchmark by year and overlap bucket. This analysis shows that temporal OOD is not simply "harder cold-start." Instead, the gap is concentrated in specific year bands and overlap regimes, which helps explain why a method that looks strong under one split can lose under another. We further include matched 3-seed support on `BindingDB_Ki` and `DAVIS` to test whether the same external panel remains stable across shifted target-cold settings. Recent molecular benchmark studies have made analogous arguments about OOD sensitivity in adjacent tasks [9, 10, 11]; our contribution is to show the same issue concretely in DTI and to release the benchmark suite as a reusable split-plus-evaluator resource for reliable DTI evaluation.

Our main contributions are:

1. We construct a matched 3-seed synthetic reference panel on `BindingDB_Kd` and show that even standard cold-start evaluation is regime-dependent.
2. We establish promoted temporal/provenance benchmarks on patent and non-patent BindingDB slices and show that their ranking can differ sharply from the synthetic panel.
3. We expand the model panel beyond pooled-LM baselines with a learner-control `RF` system and a graph-family `GraphDTA-style` probe, showing that benchmark sensitivity is not confined to one architecture family.
4. We provide year-band and overlap-bucket diagnoses, matched support on `BindingDB_Ki` and `DAVIS`, and reusable split manifests and evaluator resources to make the benchmark suite easy to inspect and reuse.

The rest of the paper first defines the benchmark setup, then reports the synthetic and temporal comparisons, and finally discusses why benchmark-qualified reporting is necessary for DTI generalization claims.

2 Related Work

Synthetic cold-start evaluation in DTI. Cold-start evaluation has become standard practice in DTI and drug-target affinity prediction. Recent models routinely report random, held-out drug, held-out target, or blind-start splits on BindingDB-derived and DAVIS-derived benchmarks [3, 4, 5, 6, 7, 8]. This literature established synthetic cold-start testing as a useful proxy for inductive generalization. However, it often treats different split definitions as variations of the same evaluation problem. Our results suggest that this assumption is too coarse, because different synthetic regimes already produce different model rankings.

Retrieval, inductive, and memory-based DTI models. Another line of work designs architectures intended to transfer information from seen entities to unseen drugs or targets. Retrieval-based or transfer-oriented models attempt to reuse evidence from related compounds and proteins, while inductive or prototype-like systems aim to improve robustness under entity holdout [4, 5, 6, 7, 8]. Our work does not claim that these mechanisms are ineffective. On the contrary, we recover setting-specific gains for retrieval on synthetic `blind_start` and for target-memory on synthetic `unseen_target`. The central point is different: benchmark choice determines whether these gains appear.

Distribution shift and benchmark design. Recent benchmark studies in molecular machine learning and related biomedical prediction problems have shown that model rankings can change under scaffold, temporal, or domain-shifted evaluation [9, 10, 11]. Our paper brings that diagnostic perspective into DTI. Unlike a leaderboard paper, we intentionally keep the model panel compact. The main result is not that one architecture dominates many baselines, but that even a small, representative panel is enough to expose ranking instability across benchmark definitions.

Positioning. Prior DTI papers typically optimize for one split regime or propose a stronger model within a fixed benchmark setup. This paper instead studies whether model rankings remain stable across synthetic cold-start regimes and a real-OOD patent benchmark, then diagnoses the shift structure behind disagreements.

3 Benchmark Setup

Our goal is to test whether an expanded DTI model panel preserves its ranking across synthetic cold-start regimes and real distribution shifts. The panel used in the revised manuscript contains five models. `base` is a fixed-feature MLP reference system built from RDKit Morgan fingerprints and hashed protein k-mer features. `RF` is a learner-control `RandomForest` classifier trained on the exact same handcrafted pair representation as `base`, allowing us to separate the effect of the learner from the effect of the representation. `DTI-LM` is a pooled ChemBERTa+ESM2 baseline aligned to recent literature, `HyperPCM` is a task-conditioned pooled-LM baseline, and `GraphDTA-style` is a graph-family probe with a molecular GCN and protein CNN. We intentionally keep the panel modest rather than exhaustive, but it now spans fixed-feature neural, fixed-feature classical, pooled-LM, and graph-based inductive biases. Because the anchor system is central to every ranking claim, Appendix Table A3 gives the exact `base` feature construction, architecture, parameter count, and training recipe.

We evaluate this panel on two benchmark groups. The first is a synthetic reference panel based on `BindingDB_Kd`, with three split definitions: `unseen_drug`, `blind_start`, and `unseen_target`. The second is a real-OOD screening panel. Our primary real-OOD benchmark is the BindingDB patent split (`patent_temporal`), which uses a year-based temporal partition and is evaluated over three seeds. We also include `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting shifted target-cold benchmarks that test whether the same external model family remains stable across assay and dataset shifts. Table 1 summarizes the shift taxonomy and split statistics.

We report AUPRC as the primary metric and AUROC as a secondary metric. AUPRC is emphasized because class balance and ranking behavior are central to the benchmark claims in this paper. For the expanded panel, we report matched three-seed results on the synthetic reference panel and the promoted temporal benchmarks. We first run transparent seed0 screening for the newly added baselines, then promote them where the screening result is either strong enough to affect the panel story or informative enough to test a reviewer concern. `RF` is promoted on all main and temporal axes because its seed0 performance is unexpectedly strong across the board. `GraphDTA-style` is promoted on `unseen_drug`, `unseen_target`,

Table 1: Benchmark taxonomy and seed0 split statistics. Test prevalence differs substantially across benchmark families, so AUPRC should be compared within benchmark rather than across rows.

Benchmark	Shift family	Synth.?	Test rows	Pos. rate
`Kd / unseen_drug`	drug-cold	yes	12,744	0.1966
`Kd / blind_start`	drug+target-cold	yes	2,821	0.2031
`Kd / unseen_target`	target-cold	yes	12,052	0.2239
`patent / patent_temporal`	temporal+source	no	49,028	0.5745
`Ki / unseen_target`	assay+target-cold	mixed	70,380	0.4652
`DAVIS / unseen_target`	dataset+target-cold	mixed	5,168	0.0615

`patent_temporal`, and `nonpatent_temporal` because these are the axes most relevant to the benchmark claims and to the request for a graph-based family. Appendix Table A8 and the baseline-expansion report make this promotion logic explicit. We further add paired bootstrap uncertainty support for the headline expanded-panel comparisons, especially those involving `RF` on synthetic drug-cold, blind-start, and the temporal benchmarks. Appendix Table A7 collects these intervals.

To make the benchmark reusable, each split is exported as a canonical resource bundle with manifests, split tables, cached identifiers, and a standalone CSV-based evaluator. This resource layer allows external predictions to be scored against the exact same test partitions used in the paper. Appendix Table A9 enumerates the released artifacts and how they are consumed by the evaluator.

The patent benchmark requires several setup choices that are documented explicitly. We use a local year-based split of `BindingDB_patent`, with training years before the validation year, a held-out validation year, and a later temporal test set. Because this benchmark also changes provenance and assay composition, we describe it as temporal/provenance OOD rather than pure time shift. We also use a patent-specific affinity conversion rather than reusing the standard `pKd` transform, because the patent benchmark stores affinities in a different log-scale convention. Appendix Table A1 summarizes the exact year boundaries, split counts, label construction, overlap policy, and seed semantics for both `patent_temporal` and the alternative-cutoff `patent_temporal_v2017`. Appendix Table A2 provides the same information for the review-driven `BindingDB_nonpatent_Kd / nonpatent_temporal` probe. BindingDB-derived panels are anchored in the BindingDB resource [1], while DAVIS provides an external kinase-focused support benchmark [2].

Finally, for diagnostic analysis, we use the saved overlap score from each test prediction to compute overlap buckets that summarize how strongly each test example resembles the training support. Concretely, `overlap_score` is defined as $\min(s_d, s_t)$, where s_d is the cosine similarity between the test drug's Morgan fingerprint and its nearest training-drug neighbor, and s_t is the cosine similarity between the test target's hashed k-mer feature vector and its nearest training-target neighbor. We use the minimum to conservatively define overlap: a pair is high-overlap only when both its drug and target are well represented in the training set. We retain cosine for the released overlap score so that drug-side and target-side overlap share the same similarity scale inside this joint diagnostic; the separate synthetic similarity analysis still reports the standard Morgan-fingerprint Tanimoto and scaffold-overlap statistics. We report benchmark taxonomy and split statistics, including test-set size and prevalence, so that AUPRC is interpreted within benchmark rather than across benchmark families.

Table 2: Synthetic reference panel and promoted real-OOD benchmarks under the expanded five-model panel. Values are mean ± std over seeds. Best mean AUPRC in each row is bold. The classical learner-control baseline (`RF`) is trained on the same Morgan+k-mer inputs as `base`; `GraphDTA-style` provides the architecture-diverse graph-family probe.

Benchmark	Seeds	base (AUPRC/AUROC)	RF (AUPRC/AUROC)	DTI-LM (AUPRC/AUROC)	HyperPCM (AUPRC/AUROC)	GraphDTA-style (AUPRC/AUROC)	Note	Winner
`BindingDB_Kd / unseen_drug`	3	0.5509 ± 0.0900 / 0.7794 ± 0.0534	**0.7339 ± 0.0051 / 0.8854 ± 0.0088**	0.5004 ± 0.0341 / 0.7364 ± 0.0122	0.5694 ± 0.0473 / 0.7778 ± 0.0448	0.5626 ± 0.0172 / 0.7988 ± 0.0129	adding a learner-control baseline rewrites the synthetic drug-cold ordering	RF
`BindingDB_Kd / blind_start`	3	0.3737 ± 0.0229 / 0.6778 ± 0.0151	**0.4512 ± 0.0373 / 0.7451 ± 0.0260**	0.3636 ± 0.0693 / 0.6490 ± 0.0642	0.3828 ± 0.1024 / 0.6807 ± 0.0824	–	the earlier "no reliable winner" row becomes RF-favored in the expanded panel	RF
`BindingDB_Kd / unseen_target`	3	**0.5224 ± 0.0602 / 0.7741 ± 0.0161**	0.5183 ± 0.0612 / 0.7838 ± 0.0218	0.5136 ± 0.0730 / 0.7970 ± 0.0159	0.4955 ± 0.0793 / 0.7810 ± 0.0191	0.4722 ± 0.0925 / 0.7637 ± 0.0314	synthetic target-cold reverts to `base` on mean AUPRC	base
`BindingDB_patent / patent_temporal`	3	0.7772 ± 0.0086 / 0.7223 ± 0.0109	**0.8234 ± 0.0012 / 0.7773 ± 0.0017**	0.7825 ± 0.0078 / 0.7362 ± 0.0083	0.7753 ± 0.0050 / 0.7368 ± 0.0090	0.7726 ± 0.0016 / 0.7232 ± 0.0007	temporal/provenance OOD changes again once the learner-control baseline is added	RF
`BindingDB_nonpatent_Kd / nonpatent_temporal`	3	0.6369 ± 0.0173 / 0.7333 ± 0.0081	**0.7201 ± 0.0012 / 0.7627 ± 0.0020**	0.6531 ± 0.0066 / 0.7644 ± 0.0041	–	0.6635 ± 0.0142 / 0.7334 ± 0.0096	the non-patent temporal probe confirms that the expanded panel is not LM-dominated	RF

4 Results

4.1 Synthetic cold splits are already regime-dependent

We begin with the `BindingDB_Kd` synthetic reference panel. Table 2 shows the main pattern: even within a single dataset, model rankings are not stable across split definitions, and the ranking changes further once the learner-control and graph-family additions are made. On `unseen_drug`, the strongest model is no longer `HyperPCM` but `RF`, which reaches 0.7339 ± 0.0051 versus 0.5509 ± 0.0900 for `base`, 0.5694 ± 0.0473 for `HyperPCM`, 0.5004 ± 0.0341 for `DTI-LM`, and 0.5626 ± 0.0172 for `GraphDTA-style`. The paired bootstrap interval for `RF - base` is strictly positive, and `RF` also remains above `HyperPCM` under paired bootstrap. On `blind_start`, the earlier "no reliable winner" result disappears once `RF` is added: `RF` reaches 0.4512 ± 0.0373, compared with 0.3737 ± 0.0229 for `base`, and the paired bootstrap interval for `RF - base` remains strictly positive. On `unseen_target`, however, the synthetic target-cold story is more stable. `base` remains strongest by mean AUPRC at 0.5224 ± 0.0602, while `RF` is close but slightly lower at 0.5183 ± 0.0612, and the paired bootstrap interval for `RF - base` crosses zero.

These results matter because they show that "synthetic cold-start DTI" is already not a single evaluation condition. Different synthetic regimes reward different model families, and the expanded panel shows that this is not merely a pooled-LM phenomenon. A classical learner on exactly the same handcrafted drug/target representation as `base` can become the strongest model on two synthetic axes, while the graph-family probe remains competitive but does not dominate. The review-driven similarity diagnosis sharpens this point further: on synthetic `unseen_drug`, the held-out compounds still have median nearest-train Tanimoto 0.7732 and Murcko scaffold match rate 0.6204. When we replace entity holdout with a stricter scaffold split, the earlier `HyperPCM` edge is no longer supported. Over three seeds on `BindingDB_Kd / scaffold_drug`, `base` reaches $0.6069 \pm 0.0161/0.8351 \pm 0.0037$, while `HyperPCM` reaches $0.5983 \pm 0.0368/0.8155 \pm 0.0204$, and the paired bootstrap interval for `HyperPCM - base` crosses zero (Appendix Tables A6 and A7). The synthetic panel therefore remains useful, but both its drug-cold ordering and its apparent best learner are sensitive to how chemically strict the split actually is.

4.2 Real temporal shift favors a different learner again

We next evaluate the expanded panel on the promoted temporal benchmarks. Once `RF` is added, the real-OOD picture changes again. On `BindingDB_patent / patent_temporal`, `RF` becomes the strongest model by mean AUPRC, reaching 0.8234 ± 0.0012, compared with 0.7772 ± 0.0086 for `base`, 0.7825 ± 0.0078 for `DTI-LM`, 0.7753 ± 0.0050 for `HyperPCM`, and 0.7726 ± 0.0016 for `GraphDTA-style`. Figure 1 therefore needs a stronger reading than in the earlier four-model draft: synthetic cold splits favor different systems, but the promoted temporal benchmarks now favor a classical learner-control baseline rather than a pooled-LM system. This advantage is statistically strong. Paired bootstrap gives `RF - base` $= +0.0463$ with 95% CI $[+0.0441, +0.0484]$ and `RF - DTI-LM` $= +0.0409$ with 95% CI $[+0.0384, +0.0433]$.

A second temporal robustness probe strengthens this reading further. On the non-patent benchmark built directly from the official BindingDB release, `RF` again becomes the strongest model over matched 3-seed evaluation, reaching $0.7201 \pm 0.0012/0.7627 \pm 0.0020$, versus $0.6531 \pm 0.0066/0.7644 \pm 0.0041$ for `DTI-LM`, $0.6369 \pm 0.0173/0.7333 \pm 0.0081$ for `base`, and $0.6635 \pm 0.0142/0.7334 \pm 0.0096$ for

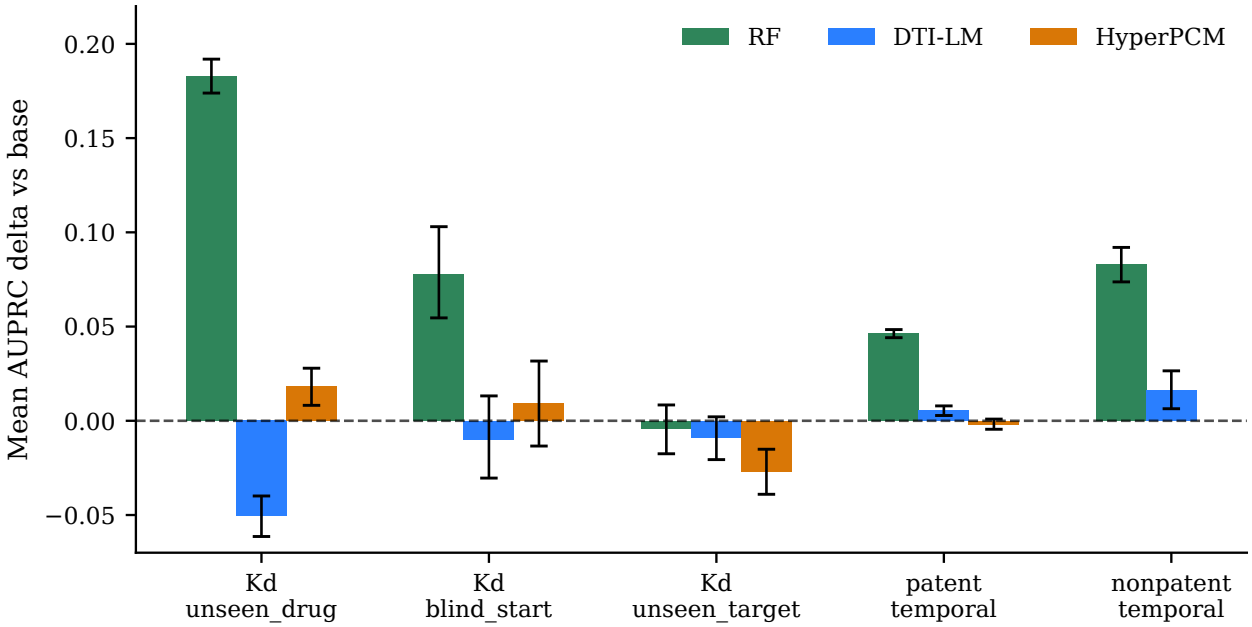

Figure 1: Mean within-benchmark AUPRC delta relative to `base` under the expanded panel. Bars are shown for the compared systems with paired-bootstrap support currently available. Because prevalence differs across benchmarks, the vertical scale should be read as a row-wise within-benchmark contrast rather than as a cross-benchmark effect-size scale.

`GraphDTA-style`. Here too, the paired bootstrap intervals stay strictly positive for both `RF - base` and `RF - DTI-LM`. These additions change the scientific read of the temporal axis. The earlier conclusion that "pooled-LM features are best on temporal OOD" no longer survives the expanded panel. The stronger and more defensible conclusion is that temporal benchmarks do not preserve the ranking seen on synthetic panels, and that a classical learner on fixed handcrafted inputs can dominate both synthetic and temporal regimes.

At the same time, two controlled patent variants show that this temporal ordering is not immutable once overlap is constrained more aggressively. When we remove recurrent train pairs (`patent_temporal_pair_novel`), `DTI-LM` becomes the strongest AUPRC model at 0.7491 ± 0.0076, ahead of `RF` at 0.7393 ± 0.0015 and `base` at 0.7378 ± 0.0107. The paired bootstrap interval for `DTI-LM - RF` is strictly positive, while `RF - base` crosses zero. When we remove seen-drug overlap (`patent_temporal_drug_novel`), `DTI-LM` again remains strongest by AUPRC at 0.7402 ± 0.0077, versus 0.7356 ± 0.0016 for `RF`, with a paired interval for `DTI-LM - RF` that also stays positive. These review-driven controls make the temporal conclusion more disciplined rather than weaker: the promoted patent axis is sensitive not only to benchmark family but also to how much entity or pair recurrence is allowed inside a nominally temporal split.

This is the central revised result. The synthetic panel does not produce a single best model, and the promoted temporal benchmarks do not preserve the smaller panel's earlier ordering either. Instead, both benchmark axis and model family change the apparent ranking within the studied panel. The alternative-cutoff benchmark from the earlier revision remains useful because it still shows that temporal ordering is not an artifact of a single year boundary, but the expanded five-model panel and the controlled patent variants now make the broader point more strongly: the apparent best system can change not only across benchmarks but also across overlap policies inside the same temporal family.

4.3 Temporal shift is structured across year bands and overlap levels

To help explain why the temporal/provenance benchmark changes the ranking, we analyze the patent results by year band and by overlap bucket. Table 3 makes the primary AUPRC slices explicit, while Figure 2

Table 3: Compact AUPRC summary behind the patent diagnosis under the expanded panel. Values are mean ± seed std over the matched 3-seed runs. The tiny 2021 slice is pooled into the `2020-2021` stratum to avoid over-interpreting a 72-example cell.

Slice type	Slice	Count	base	RF	DTI-LM
Year	2019	42,009	0.7626 ± 0.0098	**0.8062 ± 0.0013**	0.7734 ± 0.0071
Year	2020–2021	7,019	0.8716 ± 0.0121	**0.9098 ± 0.0005**	0.8533 ± 0.0100
Overlap bucket	0 (0.000–0.531)	16,341	0.5551 ± 0.0190	0.6005 ± 0.0034	**0.6353 ± 0.0175**
Overlap bucket	1 (0.531–0.710)	16,342	0.7218 ± 0.0167	**0.7606 ± 0.0012**	0.7489 ± 0.0094
Overlap bucket	2 (0.710–0.989)	16,345	0.9237 ± 0.0030	**0.9426 ± 0.0005**	0.8866 ± 0.0097

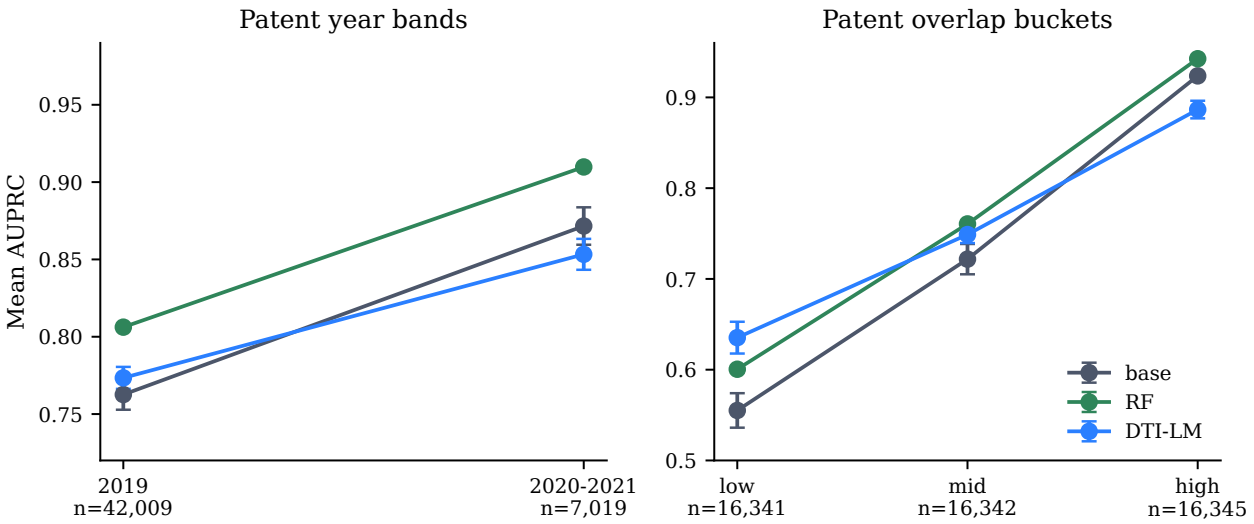

Figure 2: Patent temporal/provenance diagnosis in terms of mean AUPRC under the expanded panel. Left: pooled year-band slices, with 2021 merged into the recent stratum. Right: test-set overlap buckets computed from the saved overlap score. Error bars denote seed std over the matched 3-seed runs. In the expanded panel, `RF` dominates the aggregate temporal benchmark, while `DTI-LM` remains strongest only in the lowest-overlap bucket.

visualizes the same structure. To avoid over-reading the tiny 2021 cell, we pool 2020 and 2021 into a single recent stratum. Under the expanded panel, `RF` is strongest in both the large 2019 slice and the pooled 2020–2021 slice, so the year-band view no longer supports a simple "LM wins temporal OOD" interpretation. Instead, it shows that the aggregate temporal ranking is not tied to one model family alone.

The overlap-bucket analysis reveals another useful pattern. In the lowest-overlap bucket, `DTI-LM` still remains the strongest system, but `RF` becomes strongest in the middle- and high-overlap buckets and therefore dominates the aggregate temporal benchmark. The expanded panel thus strengthens, rather than weakens, the structured-shift diagnosis: temporal OOD is not just a colder version of synthetic cold-start, because different overlap strata continue to favor different model families. The benchmark outcome depends on both the shift axis and the modeling family, not only on whether a model uses pooled LM features.

4.4 Supporting target-cold OOD benchmarks still favor the plain base model

We treat `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting real-OOD benchmarks rather than primary headline axes. Under matched three-seed evaluation, neither promoted external baseline overturns the support-panel story. On `BindingDB_Ki`, `base` reaches $0.6934 \pm 0.0257/0.7192 \pm 0.0068$, compared with $0.6648 \pm 0.0177/0.6936 \pm 0.0044$ for `DTI-LM` and $0.6515 \pm$

$0.0301/0.6757 \pm 0.0211$ for HyperPCM. On DAVIS, base reaches $0.4097 \pm 0.0275/0.8613 \pm 0.0153$, compared with $0.4090 \pm 0.0570/0.8718 \pm 0.0152$ for DTI-LM and $0.3840 \pm 0.0269/0.8610 \pm 0.0245$ for HyperPCM.

The DAVIS result is the closest supporting-panel case: DTI-LM is effectively tied with base on mean AUPRC and slightly higher on mean AUROC, but it still does not create a clean ordering separation of the kind seen on the promoted patent benchmark. BindingDB_Ki is more decisive, with base ahead of both external baselines on both metrics. These auxiliary results are therefore consistent with the main claim of the paper, but they do not carry the same argumentative weight as the patent benchmark because they do not add a cleaner shift axis than temporal/provenance OOD.

5 Discussion

The main message of this study is that, within the studied expanded panel, DTI evaluation is benchmark-sensitive in a way that current reporting practice often obscures. A synthetic split can still be useful, but it should not be treated as a universal proxy for real OOD generalization. Once the panel includes pooled-LM baselines, a graph-family probe, and a learner-control classical baseline on the same handcrafted inputs as base, the sign and magnitude of an apparent gain can still change when the benchmark changes.

An important design choice in this paper is the use of a deliberately expanded but still finite external model panel. A larger model zoo could be added later, but the present panel is already broad enough to invalidate the narrow objection that the earlier ranking changes were artifacts of comparing only pooled-LM variants. In particular, RF and base share the same handcrafted inputs but use different learners, while GraphDTA-style adds a graph-family inductive bias. The new results therefore support a stronger statement than in the smaller panel: benchmark-qualified ranking changes persist even when the comparison spans classical, fixed-feature neural, pooled-LM, and graph-based systems. At the same time, the present paper should still not be read as an exhaustive field-wide census.

The paper also has clear limitations. Although the support benchmarks BindingDB_Ki and DAVIS remain useful counterexamples, they are still supporting rather than co-equal headline axes. The expanded model family is much stronger than the original in-repo probe panel, but it is still partial rather than exhaustive. We now add two targeted robustness probes, and both make the manuscript more disciplined. The non-patent temporal benchmark shows that the temporal ordering is not confined to patent provenance alone, while the scaffold-drug probe shows that the original synthetic unseen_drug split was chemically permissive enough that the earlier HyperPCM edge does not survive a stricter holdout. Even with these additions, the patent benchmark is still temporal/provenance OOD rather than a pure time-only shift, so the current paper should not claim that time alone explains the observed ranking change. A further caveat is that ChemBERTa and ESM2 were pretrained on broad corpora that may already include molecular and protein entities appearing in our temporal test sets. We therefore cannot rule out that part of the earlier DTI-LM edge on temporal benchmarks reflects broader pretraining coverage of chemical or protein space rather than temporal robustness per se. At the same time, the expanded panel now shows that the main temporal winner is no longer a pooled-LM model but a classical learner-control baseline using the same handcrafted representation as base. This weakens any mechanistic story that rests only on LM pretraining and shifts the more credible conclusion toward benchmark sensitivity and learner sensitivity. We also do not evaluate downstream biological decision quality. This paper should therefore be read as a well-supported benchmark case study rather than an exhaustive census of all real-OOD DTI settings. These limitations should be stated plainly, but they do not weaken the more disciplined central finding that synthetic and real-OOD rankings can disagree sharply in the tested expanded panel.

6 Conclusion

We presented a benchmark-first study of distribution shift in DTI prediction. In the expanded five-model panel, a 3-seed synthetic reference panel on `BindingDB_Kd` showed that cold-start regimes are already heterogeneous: `RF` is strongest on synthetic `unseen_drug` and `blind_start`, while `base` remains strongest on synthetic `unseen_target` by mean AUPRC. Review-driven robustness checks then made the split semantics sharper: a scaffold-drug probe removed the earlier `HyperPCM` edge, and a non-patent temporal probe showed that the expanded temporal ordering is not confined to the patent benchmark alone. Under matched 3-seed evaluation on both `BindingDB_patent` and `BindingDB_nonpatent_Kd`, the strongest model is `RF`, while the `GraphDTA-style` graph-family probe remains competitive but not dominant. Pooled patent year-band and overlap-bucket analyses further showed that real OOD is structured rather than monolithic. Overall, these results indicate that, within the studied expanded panel and tested benchmarks, synthetic cold splits and real distribution shifts should not be treated as interchangeable evaluation settings, and that learner choice can matter as much as representation family. This manuscript should be read as a benchmark/resource case study rather than an exhaustive field-wide census. Future DTI papers should therefore report both synthetic and real-OOD evidence when making generalization claims.

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Table A1: Explicit specification of the promoted patent benchmarks. The temporal partition is fixed by year; the random seed changes model stochasticity, not the partition itself.

Aspect	`patent_temporal`	`patent_temporal_v2017`
Source files	`train_val.csv + test.csv`	`train_val.csv + test.csv`
Train years	2013–2017	2013–2016
Validation year	2018	2017
Test years	2019–2021	2018–2021
Rows (train / valid / test)	142,275 / 41,155 / 49,028	90,771 / 51,504 / 90,183
Pos. rate (train / valid / test)	0.566 / 0.534 / 0.575	0.568 / 0.562 / 0.556
Label transform	raw patent value $Y \mapsto \mathrm{p}K_d = 9 - Y/\ln(10)$; positive label if $\mathrm{p}K_d \geq 7.0$	
Overlap policy	no cold-entity filtering; entity overlap is allowed, exported, and diagnosed via `overlap_score` buckets	
Duplicate policy	pair rows are kept after year partition; entity export tables are deduplicated by canonical ID	
Seed semantics	partition is fixed across seeds; seed affects training / initialization only	
OOD interpretation	temporal/provenance OOD rather than pure time-only shift	

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A Appendix

A.1 Patent benchmark specification

Because the main paper depends on `BindingDB_patent / patent_temporal`, we make its construction explicit here. Table A1 enumerates the year boundaries, split counts, label construction, overlap policy, and seed semantics for both the main patent split and the earlier-cutoff robustness variant.

A.2 Non-patent temporal probe specification

Because the main revision also relies on `BindingDB_nonpatent_Kd / nonpatent_temporal` to separate temporal drift from patent provenance, we make that construction explicit as well. Table A2 lists the source file, year cutoffs, label transform, prevalence, duplicate policy, and seed semantics for the released non-patent temporal split.

Table A2: Explicit specification of the non-patent temporal probe. This benchmark is built from the official BindingDB TSV after filtering to non-patent, positive-K_d, single-chain rows with a valid publication year. The partition is fixed by publication year; the random seed changes model stochasticity, not the split.

Aspect	`nonpatent_temporal`
Source file	official `BindingDB_All_202603_tsv.zip`, filtered to empty `Patent Number`
Date field	publication year from `Date of publication`
Train years	1977–2018
Validation year	2019
Test years	2020–2023
Rows (train / valid / test)	35,755 / 3,006 / 7,174
Pos. rate (train / valid / test)	0.341 / 0.445 / 0.426
Label transform	raw K_d in nM with $\mathrm{p}K_d = -\log_{10}(Y/10^9)$; positive label if $\mathrm{p}K_d \geq 7.0$
Overlap policy	no cold-entity filtering; `drug_overlap`, `target_overlap`, and `overlap_score` are exported for diagnosis
Duplicate policy	exact duplicate rows are removed before partitioning; no exact duplicates remain in the released split files
Seed semantics	partition is fixed across seeds; seed affects training / initialization only
OOD interpretation	non-patent temporal shift; cleaner than patent provenance, but still not a pure time-only intervention

Table A3: Specification of the `base` reference model used throughout the benchmark. This table is included to make the anchor system reproducible rather than treating it as an informal baseline label.

Component	Specification
Drug input	RDKit Morgan fingerprint, radius 2, 2048 bits, generated from canonical SMILES in `src/raicd/featurizers.py`.
Target input	L2-normalized hashed amino-acid 1/2/3-mer count vector, 2048 dimensions, generated in `src/raicd/featurizers.py`.
Drug encoder	Two-layer MLP: $\mathrm{Linear}(2048, 512)$, GELU, dropout 0.1, $\mathrm{Linear}(512, 256)$, followed by L2 normalization.
Target encoder	Same architecture as the drug encoder, with independent parameters.
Pair head	Concatenate $[d,\ t,\ d \odot t,\ \lvert d - t \rvert] \in \mathbb{R}^{1024}$, then $\mathrm{Linear}(1024, 512)$, GELU, dropout 0.1, $\mathrm{Linear}(512, 1)$.
Trainable parameters	2,886,145 parameters in total.
Optimization	BCE-with-logits objective, Adam with learning rate 10^{-3} and weight decay 10^{-4}, 8 epochs, batch size 256.
Labeling	Binary label derived from a `pKd` threshold of 7.0 (or patent-specific log-nM conversion for `BindingDB_patent`; see Table A1).
Code path	Model implementation in `src/raicd/model.py`; training entry point in `scripts/train_raicd.py`.

A.3 Reference-model specification

Because every ranking claim is anchored against `base`, we make that system explicit rather than treating it as an unnamed baseline. Table A3 specifies the exact input features, network structure, parameter count, optimization recipe, and implementation path used in every experiment.

A.4 Promoted learner-control and graph-family specifications

Because the expanded panel now hinges on the `RF` learner-control baseline and the `GraphDTA-style` graph-family probe, we make those systems explicit as well. Table A4 mirrors the level of detail used for `base`, and also records the core `RF` hyperparameters that produced the promoted results.

Table A4: Specification of the promoted learner-control and graph-family additions. This table mirrors Table A3 so the new baselines can be reproduced as precisely as the `base` anchor.

Component	Specification
GraphDTA-style drug input	RDKit molecular graph built from canonical SMILES. Node features concatenate atom symbol, degree, hydrogen count, valence, formal charge, aromaticity, and hybridization indicators in `src/raicd/graph_baselines.py`.
GraphDTA-style drug encoder	Three-layer GCN: $\mathrm{GCNConv}(36, 128)$, $\mathrm{GCNConv}(128, 128)$, $\mathrm{GCNConv}(128, 128)$, each followed by ReLU and dropout 0.1. Global mean and global max pooling are concatenated to produce a 256-dimensional drug embedding.
GraphDTA-style target input	Amino-acid token sequence with vocabulary size 21 (20 residues + padding), truncated or zero-padded to length 1000.
GraphDTA-style target encoder	Embedding dimension 64, followed by three 1D convolutions $\mathrm{Conv1d}(64, 128, k = 8)$, $\mathrm{Conv1d}(128, 128, k = 8)$, $\mathrm{Conv1d}(128, 128, k = 8)$, each followed by ReLU and dropout 0.1. Global max pooling produces a 128-dimensional target embedding.
GraphDTA-style pair head	Concatenate the 256-dimensional drug embedding and 128-dimensional target embedding, then apply $\mathrm{Linear}(384, 256)$, ReLU, dropout 0.1, $\mathrm{Linear}(256, 128)$, ReLU, dropout 0.1, $\mathrm{Linear}(128, 1)$.
GraphDTA-style trainable parameters	499,009 parameters in total under the default promoted configuration.
GraphDTA-style optimization	BCE-with-logits objective with train-split positive-class weighting, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, 8 epochs, batch size 128.
RF core hyperparameters	`RandomForestClassifier` with `n_estimators=` 400, `max_depth`=None, `min_samples_leaf=` 1, `max_features=sqrt`, `class_weight=balanced_subsample`, and seed-specific `random_state`.
Code path	Graph probe implementation in `src/raicd/graph_baselines.py` with training in `scripts/train_graph_baseline.py`; learner-control baseline implementation in `src/raicd/classical_baselines.py` with training in `scripts/train_classical_baseline.py`.

Table A5: Secondary real-OOD support benchmarks under the expanded panel. These panels use matched 3-seed evaluation, but remain supporting evidence because they reinforce rather than replace the temporal benchmark story established in the main panel. To keep promotion focused on the decision-relevant synthetic and temporal axes, the support table reports the promoted external baselines rather than the full five-model panel.

Benchmark	base (AUPRC/AUROC)	DTI-LM (AUPRC/AUROC)	HyperPCM (AUPRC/AUROC)	Winner
`BindingDB_Ki / unseen_target`	**0.6934** $\pm$ **0.0257 / 0.7192** $\pm$ **0.0068**	0.6648 $\pm$ 0.0177 / 0.6936 $\pm$ 0.0044	0.6515 $\pm$ 0.0301 / 0.6757 $\pm$ 0.0211	base
`DAVIS / unseen_target`	**0.4097** $\pm$ **0.0275 / 0.8613** $\pm$ **0.0153**	0.4090 $\pm$ 0.0570 / 0.8718 $\pm$ 0.0152	0.3840 $\pm$ 0.0269 / 0.8610 $\pm$ 0.0245	base

A.5 Additional real-OOD support benchmarks

The main paper focuses on the strongest evidence chain: the synthetic `BindingDB_Kd` reference panel, the promoted temporal benchmarks, and the patent year/overlap diagnosis. `BindingDB_Ki` and `DAVIS` remain supporting real-OOD benchmarks: both now use matched 3-seed evaluation, but they reinforce rather than replace the temporal story. To keep the promotion budget focused on the most decision-relevant axes, the support table keeps the promoted external baselines rather than rerunning the full five-model panel.

On the `BindingDB_Ki` support split, Table A5 shows that `base` remains ahead of both `DTI-LM` and `HyperPCM`. On `DAVIS`, `base` and `DTI-LM` are nearly tied on mean AUPRC, but `base` stays slightly higher on the paper's primary ranking metric. Together, these support panels reinforce the main message that strong recent baselines do not transfer uniformly across OOD definitions.

A.6 Review-driven robustness probes

The external review highlighted two concrete risks: that the promoted patent result might be driven mainly by provenance rather than temporal shift, and that the synthetic `unseen_drug` split might still be chemically too permissive. Table A6 records the targeted fixes. The first probe constructs a non-patent temporal benchmark directly from the official BindingDB release and shows that even before the learner-control

Table A6: Targeted robustness probes added after the external review. The non-patent temporal probe tests whether the promoted patent ranking persists when patent provenance is removed. The scaffold probe tests whether the synthetic `unseen_drug` ordering survives a stricter chemistry-aware holdout. The controlled patent variants reduce recurrent pair and seen-drug overlap to test how much the patent ordering depends on allowable overlap. Values are mean $\pm$ std over three seeds.

Benchmark	Systems	Mean AUPRC / AUROC	Paired AUPRC CI	Read
`BindingDB_nonpatent_Kd / nonpatent_temporal`	`base` vs `DTI-LM`	`base`: 0.6369 $\pm$ 0.0173 / 0.7333 $\pm$ 0.0081; `DTI-LM`: **0.6531 $\pm$ 0.0066 / 0.7644 $\pm$ 0.0041**	+0.0163 [+0.0064, +0.0265]	temporal ordering difference survives without patent provenance
`BindingDB_Kd / scaffold_drug`	`base` vs `HyperPCM`	`base`: **0.6069 $\pm$ 0.0161 / 0.8351 $\pm$ 0.0037**; `HyperPCM`: 0.5983 $\pm$ 0.0368 / 0.8155 $\pm$ 0.0204	−0.0086 [−0.0203, +0.0025]	stricter scaffold holdout does not preserve the original `HyperPCM` edge
`BindingDB_patent / patent_temporal_pair_novel`	`base`, `RF`, `DTI-LM`	`base`: 0.7378 $\pm$ 0.0107 / 0.6813 $\pm$ 0.0115; `RF`: 0.7393 $\pm$ 0.0015 / **0.7114 $\pm$ 0.0024**; `DTI-LM`: **0.7491 $\pm$ 0.0076** / 0.7026 $\pm$ 0.0061	`DTI-LM - RF`: +0.0098 [+0.0061, +0.0134]	once recurrent train pairs are removed, `DTI-LM` overtakes `RF` on AUPRC and `RF - base` is no longer reliable
`BindingDB_patent / patent_temporal_drug_novel`	`base`, `RF`, `DTI-LM`	`base`: 0.7255 $\pm$ 0.0117 / 0.6749 $\pm$ 0.0116; `RF`: 0.7356 $\pm$ 0.0016 / **0.7116 $\pm$ 0.0022**; `DTI-LM`: **0.7402 $\pm$ 0.0077** / 0.6989 $\pm$ 0.0069	`DTI-LM - RF`: +0.0047 [+0.0009, +0.0083]	even after removing seen-drug overlap, `DTI-LM` remains strongest by AUPRC while `RF` stays above `base`

Table A7: Paired bootstrap support for the expanded-panel comparisons. Positive deltas favor the second model in the comparison; negative deltas favor the first.

Comparison	Mean AUPRC delta	95% CI
`unseen_drug`: `RF - base`	+0.1830	[+0.1739, +0.1919]
`unseen_drug`: `RF - HyperPCM`	+0.1645	[+0.1558, +0.1733]
`blind_start`: `RF - base`	+0.0775	[+0.0546, +0.1030]
`scaffold_drug`: `HyperPCM - base`	−0.0086	[−0.0203, +0.0025]
`unseen_target`: `RF - base`	−0.0042	[−0.0175, +0.0084]
`patent`: `RF - base`	+0.0463	[+0.0441, +0.0484]
`patent`: `RF - DTI-LM`	+0.0409	[+0.0384, +0.0433]
`patent_pair_novel`: `RF - base`	+0.0015	[−0.0022, +0.0049]
`patent_pair_novel`: `DTI-LM - RF`	+0.0098	[+0.0061, +0.0134]
`patent_drug_novel`: `RF - base`	+0.0101	[+0.0065, +0.0137]
`patent_drug_novel`: `DTI-LM - RF`	+0.0047	[+0.0009, +0.0083]
`nonpatent_temporal`: `DTI-LM - base`	+0.0163	[+0.0064, +0.0265]
`nonpatent_temporal`: `RF - base`	+0.0832	[+0.0737, +0.0920]
`nonpatent_temporal`: `RF - DTI-LM`	+0.0670	[+0.0564, +0.0764]

expansion, `DTI-LM` stays ahead of `base` over matched 3-seed evaluation. The second probe replaces entity holdout with a Murcko-scaffold holdout and shows that the original synthetic `HyperPCM` edge does not survive this stricter split. We further add two controlled patent variants that explicitly suppress recurrent train pairs or seen-drug overlap. These do not erase the benchmark-sensitivity finding. Instead, they sharpen it: under `patent_temporal_pair_novel` and `patent_temporal_drug_novel`, the AUPRC winner changes again, with `DTI-LM` overtaking `RF` while `RF - base` becomes unreliable on the pair-novel variant.

A.7 Paired uncertainty support

Table A7 summarizes the paired bootstrap intervals for the expanded-panel comparisons. The strongest supported deltas still include the learner-control baseline on synthetic `unseen_drug`, `blind_start`, and the aggregate patent and non-patent temporal benchmarks. At the same time, the controlled patent rows show why the paper should not overstate that temporal story. `RF - base` on `patent_pair_novel` crosses zero, while `DTI-LM - RF` is strictly positive on both controlled patent variants. By contrast, the synthetic `unseen_target` comparison `RF - base` also crosses zero, and the scaffold-split `HyperPCM - base` comparison crosses zero as well, which is exactly why those rows are described conservatively in the main text.

Table A8: Transparent seed0 screening used to decide which additional baselines were promoted to matched three-seed evaluation. The early external-baseline screen predated the non-patent temporal probe; the later panel-expansion screen added `RF` and `GraphDTA-style` under the final five-model design.

Model	patent_temporal	nonpatent_temporal	unseen_drug	blind_start	unseen_target	Promotion	Rationale
RF	0.8249 / 0.7793	0.7216 / 0.7648	0.7279 / 0.8912	0.4488 / 0.7589	0.5618 / 0.7981	main 3-seed promotion	strongest seed0 system on every screened temporal and synthetic axis except unseen_target; therefore promoted across all main axes as the learner-control baseline
GraphDTA-style	0.7741 / 0.7242	0.6673 / 0.7444	0.5386 / 0.8159	0.4245 / 0.7426	0.5289 / 0.7776	targeted 3-seed promotion	architecture-diverse graph-family probe that is competitive enough on the main synthetic and temporal axes to test whether the expanded-panel conclusion depends on fixed-vector models
DTI-LM	0.7897 / 0.7435	–	–	0.3873 / 0.7130	0.5308 / 0.8011	main 3-seed promotion	strongest seed0 recent baseline on the primary real-OOD benchmark; therefore promoted in the main text
HyperPCM	0.7686 / 0.7247	–	–	0.3896 / 0.7447	0.5294 / 0.7990	secondary 3-seed promotion	competitive on synthetic seed0 axes, but weaker on the main patent benchmark; promoted only to test whether the main conclusion changes

Table A9: Reusable benchmark artifacts released with each split. The release is designed so that external predictions can be scored without re-running the in-repo training code.

Artifact	Example files	Purpose
Split manifest	manifest.json	versioned metadata, split counts, prevalence, seed, and canonical paths
Partition tables	train.csv, valid.csv, test.csv	canonical labeled partitions with example_id and exported overlap scores
Pair tables	train_pairs.csv, test_pairs.csv	lightweight pair-only views for external model pipelines
Entity tables	drugs.csv, targets.csv	unique drug / target vocabularies aligned to the exported split IDs
Full joined table	all.csv, all_pairs.csv	complete resource snapshot for auditing or custom slicing
Standalone evaluator	scripts/evaluate_prediction_csv.py	scores a prediction CSV against the canonical reference split and returns global + overlap-bucket metrics

A.8 Expanded-panel screening and promotion

To avoid cherry-picking concerns, Table A8 summarizes the seed0 screening passes used to decide which additional baselines to promote. The early external-baseline screen prioritized the strongest recent model on the primary real-OOD benchmark `patent_temporal`; the later panel-expansion screen then added the learner-control and graph-family probes and promoted whichever systems materially changed the main synthetic or temporal read.

A.9 Reusable benchmark resources

The benchmark release is intended to be reused independently of the in-repo models. Table A9 lists the concrete artifacts exported for each split and how they interact with the standalone evaluator. In practice, an external user only needs the canonical split bundle plus a prediction CSV keyed by `example_id`; the evaluator then reproduces the same global metrics and overlap-bucket metrics reported in this paper.

A.10 Method-centric ablations

Earlier project stages explored routers, conservative target-memory variants, and selective fallback rules. These analyses were useful for method diagnosis, but they do not define the final paper narrative. The benchmark paper is intentionally centered on ranking instability across shift definitions rather than on proposing a new best-performing method.